Future-Context Requirements for Streaming Accent Conversion: A Controlled Study Using Causal Phone Translation

Dipesh Tripathi

***Abstract*—Streaming foreign accent conversion (FAC) systems choose a lookahead budget and report a single end-to-end latency, but to our knowledge prior streaming FAC work has not densely swept that budget while separately reporting algorithmic and computational latency. We train a causal phone translator in a 16-point main sweep from 0 to 640 ms, augmented with five targeted budgets around the diminishing-return region (21 in all), with every other factor held fixed, and report five principal empirical results. First, phone-translation PER decreases smoothly with lookahead at -0.045 phone error rate (PER) per doubling ($R^2 = 0.98$) with *no locatable knee*: a piecewise fit is preferred under every axis treatment we tried, yet the breakpoint moves between 40 and 180 ms and no bootstrap interval is narrower than three octaves. Instead of a knee we identify *saturation onset* at $\approx$340 ms: with the grid completed to 20 ms spacing over 200–360 ms, and against a fixed-seed repeatability floor measured from deliberate repeats ($2\sigma = 0.0029$), the $340 \rightarrow 360$ ms increment is the first to buy less than that floor, and the average per-20 ms gain stays below it thereafter. Propagating the floor's bootstrap interval moves the onset over 300–480 ms. At the 40 ms lookahead used by PHONOS's translator, our probe retains a further 45.3% relative PER reduction between 40 and 340 ms. Second, a pre-specified hypothesis that vowels and approximants would benefit more from future context than stops, fricatives, affricates and nasals is *not supported*: all seven manner classes improve by 52–72% relative, and the between-group difference is smaller than the dispersion among classes within the groups. Bootstrapping over the six test speakers rather than over phone tokens, that difference is $+0.040$ with a 95% interval of $[-0.010, +0.101]$. It still fails when errors are reweighted by a synthesised spectral-distortion proxy for acoustic cost, so this is not an artefact of PER charging every error equally. Third, canonical-phone prediction benefits more from future context than realized-phone transcription at matched capacity (-0.045 against -0.029 PER per doubling, $1.55\times$), and within the canonical-phone arm the benefit is $1.32\times$ larger in absolute PER slope at canonical-deviation positions (substitutions and deletions in the g2p–IPA alignment). Fourth, repeating the sweep on a second causalised encoder (wav2vec 2.0 base), tapped at its own selected phonetic layer, changes the answer: it starts within 0.01 PER of WavLM at L=0 yet gains only 3.9% relative from 640 ms of lookahead against WavLM's 63.3%, and where WavLM is smooth and log-linear ($R^2 = 0.99$) its response is weak and non-monotonic ($R^2 = 0.10$), bottoming out at 160 ms and rising again, a pattern reproduced at a second seed; at five times the training budget the $0/160/640$ ms ordering is unchanged and its endpoint gain shrinks to $+1.8\%$ against WavLM's $+65.8\%$. Neither the measured exchange rate nor the detailed curve shape is a universal property of the task. Fifth, per-chunk compute depends only weakly on lookahead, moving at most 1.9% on an Apple M4 and 7.6% on an ARM64 Linux host across a $17\times$ change in algorithmic latency. That is a weak empirical dependence over the tested range, subject to the throughput constraint $t_{\mathrm{cmp}} < c$. We release the harness and the per-condition outputs.**

***Index Terms*—accent conversion, streaming speech, latency, lookahead, self-supervised models**

D. Tripathi is with the University of South Dakota, Vermillion, SD, USA (e-mail: dipesh.tripathi@coyotes.usd.edu).

Harness, causality audit, sweep configurations, analysis scripts with self-tests and per-condition hypothesis dumps: https://github.com/Dipeshtripathi13/streaming-fac-lookahead. Data is L2-ARCTIC [1] via `KoelLabs/L2Arctic` (CC-BY-NC-4.0); that licence is non-commercial, so configurations are released rather than weights.

I. Introduction

Streaming FAC is established. PHONOS [2] runs at $\leq$40 ms lookahead and reports an 81% reduction in non-native accent confidence. What prior work does not provide is a dense characterisation of *why* 40 ms, what it costs in quality, or what any of it does on hardware that is not a datacentre GPU.

Three gaps motivate the study.

The budget is chosen, not characterised. PHONOS *states* 40 ms and publishes no ablation showing 40 ms is necessary or sufficient; TVTSyn [3] fixes 4 future tokens ($\approx$80 ms) without varying them. DarkStream [4] does ablate, over three points $\{0, 140, 280\}$ ms, and finds the third buys <1pp. But three points scored on anonymisation accuracy cannot tightly localise where returns stop, which is the question a deployment budget actually asks. "What would 160 ms buy you" still has no published answer.

Algorithmic and computational latency are reported fused. A figure such as PHONOS's "<241 ms on a single GPU" [2] combines a modelling choice with unstated inference and buffering costs. A reader cannot tell whether a faster chip would help. We decompose throughout as

$$t_{\mathrm{e2e}} = \underbrace{c + L}_{t_{\mathrm{alg}}} + t_{\mathrm{cmp}} + t_{\mathrm{buf}}, \tag{1}$$

where c is chunk size, L is lookahead, t_{cmp} is per-chunk compute and t_{buf} is I/O and jitter buffering. The c+L form assumes a block-emission convention: output for a chunk is produced only once the whole c ms chunk and its L ms of future context have arrived. Conventions that emit frame-synchronously, or that charge only the lookahead, give a different constant; we state ours so the terms can be re-derived under another. t_{alg} is inherent: it is the delay on an infinitely fast machine, because the model refuses to emit the frame ending at time t_n until it has observed input through t_n+L. In frames that is $N_L = \lceil L/h \rceil$ of them at hop h, and

§III-C returns to the consequences of the rounding. t_{cmp} is implementation- and hardware-dependent and can be reduced through optimization or faster hardware.

Degradation is reported in aggregate. Existing results are utterance-level MOS, WER or accentedness. We found no prior streaming FAC study reporting lookahead sensitivity by phonetic manner class.

Our contributions:

1) A latency–accuracy characterization of a causal phone-translation probe for streaming FAC across 21 lookahead budgets (a 16-point main sweep plus five targeted additions), reporting an average exchange rate and a floor-relative saturation region rather than an unidentifiable knee (§V-C).
2) A negative result on a pre-specified phoneme-class hypothesis (§V-E).
3) Evidence that canonical-phone prediction benefits more strongly from lookahead than realized-phone transcription under matched capacity, and that this holds within a single arm, with the association surviving speaker-level resampling, leave-one-speaker-out and within-phone stratification for phone identity (§V-F).
4) A cross-encoder test showing that neither the exchange rate nor the detailed curve shape transfers from WavLM base+ to wav2vec 2.0 base, replicated at a second seed (§V-D).
5) Two patches that make a masked SSL encoder causal, with truncation-based causality verification (§III-B).
6) Three documented measurement failures, each of which produced plausible output before being caught (§VI).

II. Related work

Streaming FAC is established. Nguyen et al. [5] present a streaming non-autoregressive accent-conversion system on an Emformer encoder, demonstrating streaming feasibility; PHONOS [2] subsequently targets much lower latency with bounded future context. Neither characterises how quality varies with that context, which is the axis we isolate. PHONOS, TVTSyn [3] and DarkStream [4] come from one group and pair conversion with speaker anonymisation, and two report more than is sometimes credited. TVTSyn gives CPU as well as GPU latency ($\approx$132 and $\approx$79 ms at 60 ms chunk), so a CPU number is not unpublished; a CPU latency–*quality* curve is. DarkStream ablates look-ahead over $\{0, 140, 280\}$ ms and selects 140 ms because the extra 140 ms buys $<$1pp at double the buffering delay: a three-point ablation, scored on anonymisation accuracy rather than conversion quality. DarkStream likewise reports diminishing marginal return at its largest tested budget, though a three-point ablation on a different objective is too sparse to compare saturation locations with the estimate we obtain over 21 budgets.

StreamVC [6] reports mobile operation without device, core or thread details, limiting hardware-level reproducibility. A survey [7] frames the space; streaming anonymisation [8], [9], [10] shares the constraint but not the accent objective; offline FAC [11], [12] and low-latency VC [13] bracket the design space. Our contribution is orthogonal: we characterise an axis these systems parameterise but do not study, and claim no quality improvement over any of them.

TABLE I
Truncation test on `wavlm-base-plus`. Relative L_2 change in earlier frames after deleting later audio.

configuration	relative L_2	causal?
attention mask only	1.14×10^{-2}	no
+ causal positional conv	6.00×10^{-3}	no
+ cumulative GroupNorm	2.60×10^{-2}	no
both patches	$\mathbf{4.58 \times 10^{-6}}$	**yes**

III. Method

A. A causal phone translator as the probe

Training a full FAC pipeline per lookahead is prohibitive. We instead isolate a phonetic-normalization component directly relevant to the research question; future-context needs in acoustic generation and prosody may differ. L2-ARCTIC [1] provides, per utterance, both the canonical G2P/lexicon-derived phone sequence (`g2p`) and the annotated realized phone sequence (`ipa`). We operationalize the phonetic-normalization component of accent conversion as

$$\textit{non-native audio} \rightarrow \textit{canonical phone sequence},$$

a CTC head over a frozen causal encoder. This provides a controlled supervised proxy for the phonetic-normalization stage, and swapping the target tensor from `g2p` to `ipa` yields a more tightly matched architectural control, differing in *one tensor*, than an AC-versus-VC comparison, though the two targets still differ in label entropy, sequence length and annotation noise (§V-F).

B. Making the encoder causal

Masking self-attention in wav2vec2/WavLM-base [14], [15] does *not* produce a causal encoder. Two leaks remain: the positional convolution (`pos_conv_embed`, depthwise, kernel 128, symmetric padding) admits 1.28 s of future at a 20 ms frame rate; and the feature-encoder GroupNorm normalises each channel over the entire utterance, making every output frame depend on every input frame.

We verify causality by truncation: delete audio after frame t and check whether any earlier frame moved (Table I). Note the third row: patching GroupNorm alone makes matters *worse*. Removing one leakage path does not monotonically reduce truncation sensitivity, because the remaining mechanisms interact; a partial causality fix is not a partial improvement.

The three leak magnitudes are stable across library versions and reproduce exactly. The patched row is not: at 10^{-6} it sits at float32 accumulation noise, and moves with the `transformers` release (an earlier version gave 6.14×10^{-6}). Both values are four orders of magnitude below the 10^{-4} threshold, so the verdict is unaffected, but the row should be read as “indistinguishable from causal at float32 precision” rather than as a reproducible constant.

C. *Lookahead is quantised*

Lookahead is realized as $N_L = \lceil L/h \rceil$ frames at a frame hop $h = 20\,\text{ms}$ (equivalently, a 50 Hz frame rate). A knee narrower than 20 ms is not merely unmeasured but *unrepresentable*, and sampling finer than h silently creates duplicate conditions that reweight any fit. Lookahead is therefore quantized in 20 ms increments: we sample every increment through 200 ms and a subset of quantized budgets above it, and assert zero collisions in t_{alg} across all conditions before fitting.

IV. Experimental setup

Frozen causal WavLM-base-plus, CTC phone head, 1200 steps, batch 8, chunk 40 ms, 100-frame lookback, speaker-disjoint test split of 400 utterances. Two sweeps: a 7-lookahead $\times$ 2-target $\times$ 3-seed sweep (42 runs) and a 16-lookahead $\times$ 2-seed dense sweep, run on *both* target arms (64 runs), later augmented on the canonical-phone arm with five budgets (220, 260, 300, 340, 360 ms) to resolve the diminishing-return region, all on an NVIDIA T4. Compute measurements use an Apple M4 (10 CPU cores: 4 performance, 6 efficiency; 16 GB; macOS 25.5) and an ARM64 Linux host (aarch64, 4 logical cores, 3.8 GB, Linux 6.8, scipy-openblas 0.3.29; the platform does not expose a CPU model string), CUDA-event or `perf_counter_ns` timed, 28–50 repetitions with three warm-up iterations discarded.

Repeatability floor. A conventional multi-seed design does not isolate fixed-seed run-to-run nondeterminism: pairing within seed cancels seed effects but leaves this residual, which is consistent with nondeterministic GPU kernels. We measure it directly, repeating four conditions ($L \in \{40, 160, 240, 320\}$ ms) four times each at *identical seed and configuration*, giving 12 degrees of freedom on test PER:

$$\sigma = 0.00146, \qquad 2\sigma = \mathbf{0.0029}\ \text{PER}, \tag{2}$$

with a bootstrap 95% interval of $[0.0013, 0.0034]$ on 2σ.

This supersedes an earlier estimate of $2\sigma = 0.0044$ taken from six *accidental* repeats, which the present interval excludes; that estimate was inflated by a single outlying pair. The correction matters because the saturation onset in §V-C is defined against this floor, and a smaller floor makes smaller gains measurable. We also note the floor is not constant across conditions: per-condition σ spans 0.00064 to 0.00223, a factor of 3.5. A single global threshold is therefore an approximation, and marginal decisions near it should be read as such.

V. Results

A. *Per-chunk compute depends only weakly on lookahead*

Across $L \in [0, 640]$ ms at chunk 40 ms on the M4, t_{alg} spans 40–680 ms (17$\times$) while median per-chunk compute moves from 43.23 ms to at most 44.06 ms, a range of -0.3% to $+1.9\%$. On ARM64 the span is 0 to $+7.6\%$. The two terms of Eq. 1 are therefore separately actionable, which is the practical justification for reporting them separately.

Independently actionable is not the same as free. At this configuration $t_{\text{cmp}} \approx 43$–44 ms against a 40 ms chunk, i.e. $\text{RTF} = t_{\text{cmp}}/c \approx 1.08 > 1$: the base encoder does not sustain real time at 40 ms chunks on this machine, whatever L is.

TABLE II
Convergence check: validation PER against optimization step, single seed. The ranking is preserved at all 12 checkpoints and the fitted exchange rate is stable from step 1000 onward.

step	L=0	40	160	240	640
500	0.5301	0.4375	0.3288	0.2946	0.2574
1000	0.4715	0.3601	0.2588	0.2353	0.2012
2000	0.4480	0.3392	0.2332	0.2054	0.1717
3000	0.4340	0.3293	0.2131	0.1930	0.1622
4000	0.4135	0.3152	0.1992	0.1873	0.1533
5000	0.4092	0.3108	0.1962	0.1838	0.1473
6000	0.4060	0.3100	0.1951	0.1822	0.1468

Eq. 1 describes the delay of a single chunk and must be read alongside the steady-state throughput constraint

$$\text{RTF} = t_{\text{cmp}}/c < 1, \tag{3}$$

which lookahead leaves untouched but chunk size and hardware do not. Increasing L adds little per-chunk compute over the tested range; being able to run at all is a separate constraint.

B. *Is the curve an optimization-rate curve?*

Every condition elsewhere is trained for a fixed 1200 steps, so what is measured directly is *lookahead $\rightarrow$ performance after 1200 optimization steps*. That equals *lookahead $\rightarrow$ attainable performance* only if the conditions sit at comparable points in optimization. If longer lookahead simply trained faster, part of the exchange rate would be an optimization-rate difference rather than a context effect. We test this rather than list it as a caveat.

Five representative budgets ($L \in \{0, 40, 160, 240, 640\}$ ms) were trained for 6000 steps, five times the standard budget, with validation every 500 steps, giving a 12-point trace per condition (Table II).

Three results. **Training largely flattens**: over the final 1500 steps validation PER moves by only 0.0005–0.0045 across the five conditions, the two extremes (L=0 at 0.0041, L=640 at 0.0045) still drifting slightly more than the middle three, so 6000 steps is near but not fully at convergence. We deliberately do not test these against the repeatability floor of Eq. 2: that floor estimates fixed-seed variation of *test* PER under complete reruns, which is not the same object as a validation checkpoint trajectory within a single run. **The ranking never reorders**: the ordering L=0 $> 40 > 160 > 240 > 640$ holds at **12 of 12** checkpoints, including the earliest at step 500. **The exchange rate is stable**: fitted over $L \geq 40$ it stays within -0.0396 to -0.0461 PER per doubling at every checkpoint. Validation runs every 500 steps, so this run has no checkpoint at the 1200-step budget used elsewhere; the nearest is step 1000, whose fitted slope (-0.0405) agrees with the 6000-step value (-0.0414) to within 0.0009 PER per doubling. The bracketing checkpoints agree too (-0.0461 at 500, -0.0397 at 1500), so the budget used elsewhere sits in a flat region of this curve rather than at a favourable point on it.

Improvement from step 1500 to 6000 is also roughly uniform across conditions (-0.030 to -0.045 PER) rather than concentrated at long lookahead, which is the specific

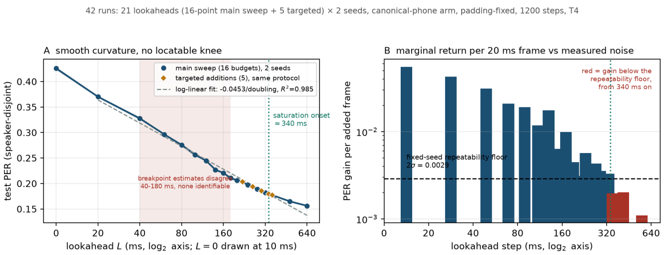

Fig. 1. Dense sweep, canonical-phone arm, 2 seeds: the 16-point main sweep (circles) with the five targeted saturation-region additions (diamonds). **Left:** smooth curvature; the shaded band spans the breakpoint estimates produced by three defensible treatments of $L=0$, none identifiable. **Right:** marginal return per added 20 ms frame against the measured 2σ floor; red bars are below it.

pattern a convergence-rate confound would produce. Absolute PER does keep falling with longer training, so the fixed-budget numbers elsewhere are a comparison at matched budget rather than converged performance. But the *relative* quantities this paper reports, the ranking and the exchange rate, are not artefacts of stopping at 1200 steps. The check is single-seed over five budgets and uses validation rather than test PER, so it bounds the concern rather than eliminating it.

C. RQ1: an exchange rate and a saturation onset, not a knee

Canonical-phone-arm PER falls monotonically from 0.4256 at $L=0$ to 0.1561 at $L=640$ ms. Fitting $L \geq 20$ ms on $\log_2 L$ gives

$$\Delta\text{PER} = -0.0452 \text{ per doubling}, \quad R^2 = 0.983, \tag{4}$$

over the 15 positive-lookahead points of the main sweep. The five budgets added later to resolve the saturation region were sampled where the curve is flattest, so we report the fit with and without them rather than silently choosing one: including all 20 positive-lookahead conditions gives -0.0453 per doubling at $R^2 = 0.985$. Targeted sampling did not move the exchange rate.

There is curvature, and it is not localisable. BIC prefers a two-segment fit under all three axis treatments we tried ($\Delta\text{BIC} = -31.0$, -40.0, -46.5 over all 21 budgets), which by the usual thresholds [16] is strong evidence. But the breakpoint lands at 40, 180 and 180 ms respectively, and the bootstrap 90% intervals span 3.2, 2.6 and 3.2 octaves ($[40, 360]$, $[60, 360]$, $[40, 360]$ ms). Twenty-one points did not rescue a knee: the five budgets added in the region where a knee would have to lie sharpened the evidence for curvature without moving the breakpoint. The reason is mechanical and worth stating: on a $\log_2(L+1)$ axis the $L=0 \to 20$ ms gap is 4.39 units while every other gap is ≈ 1.0, so a piecewise fit can be strongly biased toward placing a breakpoint just past that gap even when the underlying curve has no localised knee.

What we report instead is the lookahead beyond which marginal return falls below the measured floor. Because the original grid was 20 ms-spaced only to 200 ms, we added the five budgets 220, 260, 300, 340 and 360 ms under identical settings, making the range 200–360 ms fully 20 ms-spaced.

TABLE III
CANONICAL-PHONE-ARM PER AGAINST LOOKAHEAD (MEAN OF 2 SEEDS) AND MARGINAL RETURN PER ADDED 20 MS FRAME, IN UNITS OF THE MEASURED 2σ=0.0029 FLOOR (EQ. 2). BELOW 1.0×, THE GAIN FROM AN ADDITIONAL FRAME IS SMALLER THAN THE MEASURED FIXED-SEED REPEATABILITY THRESHOLD. THE GRID IS 20 MS-SPACED TO 360 MS; THE FINAL TWO ROWS AVERAGE OVER WIDER INTERVALS.

L (ms)	PER	gain/frame	× floor
0	0.4256	–	–
20	0.3700	0.0556	19.2
40	0.3275	0.0426	14.7
60	0.2963	0.0312	10.8
80	0.2754	0.0209	7.2
100	0.2564	0.0190	6.6
120	0.2446	0.0118	4.1
140	0.2269	0.0176	6.1
160	0.2206	0.0064	2.2
180	0.2106	0.0099	3.4
200	0.2062	0.0045	1.5
220	0.2030	0.0031	1.1
240	0.1973	0.0057	2.0
260	0.1934	0.0039	1.3
280	0.1891	0.0043	1.5
300	0.1855	0.0036	1.2
320	0.1823	0.0032	1.1
340	0.1790	0.0033	1.1
360	0.1770	0.0020	**0.7**
480	0.1649	0.0020	**0.7**
640	0.1561	0.0011	**0.4**

The quantity below is therefore a per-*observed*-step gain, not an average over wider sampled intervals.

Gains run $+0.056$ (0→20 ms), $+0.021$ (60→80) and $+0.0045$ (180→200). Against the measured floor of 0.0029 (Eq. 2), every observed 20 ms step remains above it out to 340 ms, at $1.1\times$ (200→220), 2.0, 1.3, 1.5, 1.2, 1.1 and 1.1. The observed 340 → 360 ms increment is the first to fall below it, at $0.7\times$, and the average per-20 ms gain over the two longer intervals that follow stays below it as well (0.7 over 360 → 480 and 0.4 over 480 → 640). We therefore place **saturation onset at** $\approx$**340 ms**. The wording is deliberate: above 360 ms the grid is not 20 ms spaced, so a single sub-interval could in principle exceed the floor without lifting its interval average, and what we can state is the first sustained transition rather than a proven upper bound.

This figure moves with the floor, and we report the sensitivity rather than a single number. Recomputing the onset at each end of the floor's bootstrap interval moves it from 300 ms (at $2\sigma = 0.0034$) to 480 ms (at 0.0013). This span is a sensitivity range induced by uncertainty in the floor, not a confidence interval on a fitted saturation parameter. Under the superseded 0.0044 estimate it would have been 240 ms, so the earlier figure was an artefact of an over-conservative floor, not of the curve. The gains between 200 and 340 ms sit at only 1.1–2.0× the floor, so this is a region where returns become marginal rather than a sharp cutoff. Unlike a knee, though, the quantity depends on the floor rather than on an axis choice, and the floor is now measured with 12 degrees of freedom.

The cost of a 40 ms budget. PER at L=40 ms is 0.3275; at 340 ms it is 0.1790. **Relative to the 40 ms condition, in-**

creasing lookahead to the saturation onset reduces PER by a further 45.3%, and to 640 ms by 52.3%. In absolute terms, of the 0.2466 PER separating $L{=}0$ from 340 ms, a 40 ms budget has captured 0.0981, or 39.8%. **For WavLM base+, then, 40 ms lies well before the observed diminishing-return region.** We stop at that scoped statement rather than the general claim that current budgets are under-provisioned: §V-D exhibits a backbone on which the same extra context buys almost nothing.

D. The exchange rate does not transfer across encoders

Every number above is measured over one frozen backbone, so the obvious question is whether the exchange rate describes speech or describes WavLM. We repeated the sweep on a second causalised encoder, wav2vec 2.0 base [15], holding the corpus, split, head, optimiser, step budget, seed, chunk, lookback and matched lookahead grid fixed. The pretrained backbone and the selected phonetic tap layer both differ, for reasons set out below. The matched grid is $L \in \{0, 20, 40, 80, 160, 320, 640\}$ ms, a subset of the main sweep; fitted over this reduced grid the WavLM slope is -0.0446 rather than the -0.0452 of Eq. 4, which is why Table IV differs slightly from §V-C. Both encoders are causalised by the same two patches and both pass the same truncation-based causality test.

The layer must be matched by function, not by index. Tapping wav2vec 2.0 at layer 9, WavLM's phonetic layer, produces a degenerate run: PER rises from 0.772 at $L{=}0$ to 0.928 and the training loss is still falling steeply at step 1200. Probing the stack at $L{=}160$ ms with everything else fixed shows why (PER by layer: 0.402 at 2, 0.411 at 3, 0.437 at 4, 0.586 at 6, 0.880 at 8, 0.928 at 9, 0.971 at 10). Decodability falls monotonically with depth: this encoder carries phones low in the stack, and layer 9 is seven layers past its usable representation. Matching two encoders by layer index is not matching them.

How each layer was selected, and whether the choice drives the result. The two procedures are not symmetric and we state them plainly. WavLM's layer 9 was fixed in advance from prior practice, before any lookahead data existed, and was never re-selected. wav2vec 2.0's layer was chosen by the probe above, which runs at $L{=}160$ ms, inside the sweep it is later used to interpret, so layer and lookahead sensitivity rest on partly overlapping evidence. The concern this raises is that a layer picked mid-curve might flatter or flatten the curve it is picked from. It does not here: re-selecting on PER at $L{=}0$ alone, which is the endpoint the relative gain is measured *from* rather than a point on the response, returns the same layer (0.4359 at 2, 0.4590 at 4, 0.5735 at 6, 0.6940 at 8). The comparison below is therefore insensitive to which of the two criteria is used, though it would be cleaner still to fix both layers by a pre-specified rule.

Direction replicates; magnitude does not. At the selected layer 2, wav2vec 2.0 starts within 0.01 PER of WavLM (0.4359 against 0.4256) and then barely moves: 3.9% relative gain against 63.3%, and -0.0015 PER per doubling against -0.0446. The log-linear fit that gives $R^2 = 0.989$ on WavLM gives $R^2 = 0.099$ here, and the curve bottoms out at $L{=}160$ ms before rising again. The endpoint gain nonetheless exceeds the fixed-seed repeatability floor and is reproduced at a second training seed.

TABLE IV

SAME SWEEP, SECOND ENCODER, EACH TAPPED AT ITS SELECTED PHONETIC LAYER, GIVEN IN PARENTHESES (§V-D). SLOPE IS PER PER DOUBLING OF LOOKAHEAD, FITTED OVER THE MATCHED GRID. THE TWO START WITH SIMILAR PER AT $L{=}0$ BUT EXHIBIT MARKEDLY DIFFERENT LOOKAHEAD RESPONSES.

encoder	$L{=}0$	$L{=}640$	rel. gain	slope
WavLM base+ (9)	0.4256	0.1561	+63.3%	−0.0446
wav2vec 2.0 base (2)	0.4359	0.4189	+3.9%	−0.0015
seed 7	0.4376	0.4265	+2.5%	–

A second seed reproduces the weak, non-monotonic response. The comparison carries a headline claim, so we replicated it at a second training seed on five budgets, including the 340 ms onset that the first grid did not sample. Seed 7 lands within 0.008 PER of seed 1337 at every shared budget (0.4376/0.4359 at $L{=}0$, 0.4219/0.4169 at 40, 0.4043/0.4004 at 160, 0.4265/0.4189 at 640; seed 7 gives 0.4155 at 340, where seed 1337 sampled 320 and gave 0.4028). Both curves reach their minimum at the same $L{=}160$ ms and both rise again by 640 ms, so the non-monotonicity is a reproducible feature of this encoder rather than one seed's noise. The endpoint gain is +2.5% against +3.9%: the effect is small in both, and small in the same shape.

Nor is it an artefact of the training budget. A second seed cannot rule out an optimisation effect the two seeds share: perhaps wav2vec 2.0's lookahead conditions simply separate later. We therefore give this encoder the same convergence check WavLM receives in §V-B, at $L \in \{0, 160, 640\}$ ms to 6000 steps. Every condition improves ($0.4359 \rightarrow 0.3782$, $0.4004 \rightarrow 0.3361$, $0.4189 \rightarrow 0.3714$), but they improve together, and the three-point ordering is unchanged: the minimum remains at $L{=}160$ ms and $L{=}640$ remains worse than it. Three budgets test that ordering, not the full seven-point curve. The endpoint gain does not grow with training; it *shrinks*, from +3.9% to +1.8%, while WavLM at the same 6000 steps gains +65.8% on test PER ($0.4135 \rightarrow 0.1414$). Both encoder figures are test PER and so are comparable to each other; Table II reports *validation* PER at checkpoints, on which the corresponding WavLM gain is +63.8%. Five times the budget widens the gap rather than closing it.

This is not an artefact of tapping a shallow layer. Layer 2 has had substantially less contextual mixing than deeper layers, so the natural objection is that there was less context to withhold and that sensitivity would recover deeper in the stack. We do not observe that recovery. Relative gain from $L{=}0$ to 640 ms is +3.9% at layer 2, +5.4% at 4, +5.2% at 6 and −14.1% at 8: flat across every layer where the encoder can still decode phones, then negative. We looked for the trade-off and did not find it.

Two encoders with similar zero-lookahead PER therefore differ by more than an order of magnitude in endpoint relative gain, a statement that holds for both wav2vec 2.0 seeds (+2.5% and +3.9% against +63.3%). **The measured** -0.045

PER per doubling is specific to the WavLM base+ configuration tested here, and should be read as one encoder's operating curve rather than as a fact about how much future context English phones require. The swap rules out treating the measured WavLM exchange rate as a backbone-independent law, and it also shows that the detailed shape need not carry over: WavLM's smooth diminishing-return curve has no counterpart in wav2vec 2.0's weak, non-monotonic one. We do not run the breakpoint and axis-perturbation analysis of §V-C on a curve that is not monotone, so we make no knee claim about it either way. Every positive quantity here, the exchange rate, the saturation onset and the cost attributed to a 40 ms budget, should therefore be re-measured on whatever backbone a system actually deploys. Whether the gap traces to pretraining scale (960 h against 94 k h), to WavLM's denoising and speaker-conditioned objectives, or to something else is not something a two-encoder comparison can decide.

E. RQ2: the pre-specified manner grouping does not predict lookahead benefit

We fixed this prediction in code before the hypothesis outputs it is tested on were generated. The grouping, and the rationale for it, sit in `eval/phoneme_analysis.py` at commit `c5066dc` of the public repository (2 Aug 2026),[1] seven days before the hypothesis dumps were committed (`b66d0a5`, 9 Aug 2026). We are precise about what this does and does not establish. It is not a registered pre-registration, and we do not call it one: git author and committer dates are set by the committing machine and history can be rewritten, so the chronology is a claim about our repository rather than third-party attestation. What it does give a reader is the *content* of the prediction, verbatim and inspectable, including the class assignment and the reasoning we offered for it at the time. The current public history is consistent with the prediction preceding the outputs it is tested on, but consistency is the most that can be claimed: this is not immutable third-party timestamping. The prediction was that vowels and approximants break first while stops, fricatives, affricates and nasals would survive short lookahead. The reasoning was that vocalic identity is carried by formant movement across a syllable nucleus that is tens to hundreds of milliseconds long. Peterson and Lehiste measured English syllable nuclei at roughly 80–250 ms [17], a range consistent with the vowel durations later reported over 139 talkers [18]. The second half of the reasoning was that this movement is criterial rather than incidental, since even nominal monophthongs are identified better from a trajectory than from a steady-state slice [19], with glides and diphthongs differing from vowels in rate of formant change rather than in kind [20]. Stops, fricatives and affricates were placed in the comparison group because their defining cues are comparatively localised: stop place is signalled by brief formant transitions to a locus [21] and is recoverable from a short-time spectrum at the release [22], so a narrow window should suffice. Nasals were assigned to the same group under the same short-context prediction, but the

[1] https://github.com/Dipeshtripathi13/streaming-fac-lookahead/commit/c5066dc

TABLE V

RELATIVE ERROR-RATE REDUCTION, L=0 $\rightarrow$ 640 MS. B = PRE-SPECIFIED AS BREAKING FIRST, S = AS SURVIVING. INTERVALS ARE PERCENTILE BOOTSTRAP OVER THE SIX TEST SPEAKERS, NOT OVER PHONE TOKENS. THE PREDICTED GROUPING DOES NOT ORDER THE DATA.

	class	rel. gain	95% CI	ref. phones
S	nasal	0.716	[0.668, 0.779]	4 020
B	approximant	0.700	[0.628, 0.774]	4 575
S	fricative	0.687	[0.608, 0.774]	7 659
B	vowel (mono)	0.673	[0.595, 0.767]	13 008
B	vowel (diph)	0.654	[0.577, 0.758]	4 824
S	stop	0.615	[0.550, 0.689]	7 071
S	affricate	0.525	[0.397, 0.633]	1 098

acoustic argument above is strongest for the obstruent classes and we do not claim it extends cleanly to nasals. We record this asymmetry as it stood when it was specified rather than reconstructing a stronger rationale after the grouping failed. Nasals in fact gain the most of any class (Table V). Testing this on 42,255 unique reference-phone tokens evaluated across seven lookahead conditions (295,785 phone–condition observations), with each substitution or deletion charged to the class of the *reference* phone:

Group means differ in the predicted direction (0.676 vs 0.636), but that is the weakest available reading and four stronger checks fail. The effect size is 0.66: the between-group difference (0.040) is smaller than the pooled SD across classes (0.060). The surviving group's *internal* spread is 0.190, which is 4.7× the between-group difference. And 5 of 12 pairwise orderings are violated, with nasals out-gaining all three breaks-first classes.

Under speaker-level resampling the difference is not distinguishable from zero. The 42,255 tokens per condition are not independent: they are nested in six speakers, and the claim at issue is about accented speech, for which the independent replicate is a talker rather than a phone. Resampling speakers rather than tokens (2000 draws, Table V) puts the between-group difference at $+0.040$ with a 95% interval of $[-0.010, +0.101]$, which **includes zero**. Every individual class gain remains large and clearly non-zero, with the intervals in Table V sitting between 0.40 and 0.78, so every class shows a substantial positive gain; what fails to separate from zero is the *grouping*. Token-level intervals would have been roughly an order of magnitude narrower and would have declared this difference significant, which is the error the clustering avoids. The same procedure applied to RQ3 (Sec. V-F) resolves the larger interaction there, showing that the estimator is not uniformly uninformative at six speakers, though that does not by itself establish power for an effect this small. A cluster bootstrap over six clusters leans on asymptotics that six clusters do not supply, so we add the non-asymptotic companion: recomputing the difference with each speaker removed in turn gives $+0.0217$ to $+0.0592$ (pooled $+0.0401$). No single talker drives the estimate, since the sign is stable across all six drops, but the entire leave-one-out range also sits inside an interval that contains zero, so the verdict does not change.

The pattern does not hold within L1 either, but that

split cannot carry weight. Splitting by native language, the ratio of group gains straddles unity: Vietnamese 1.23, Korean 1.19, Arabic 1.08, Spanish 1.02, Hindi 0.99, Chinese 0.92; requiring both the gain ratio and the mean slope to move in the predicted direction, only 3 of 6 qualify (Spanish clears the gain criterion but not the slope one). The reversals show that the pooled ordering is not consistent across the six held-out speakers, each of whom carries a distinct L1 label. The deeper limitation is that in the speaker-disjoint test split *each L1 is represented by exactly one speaker* (ABA, LXC, SVBI, HJK, EBVS, PNV), so L1 and talker are perfectly confounded: this table cannot separate a language effect from an individual one, and every "by-L1" row is equally readable as a single speaker's idiosyncrasy. We report it as a consistency check on the pooled result, and not as evidence about native language.

H2 is not supported. The finding is not that lookahead fails to help vowels, since every class improves 52–72% relative. The finding is that the pre-specified manner grouping does not reliably order lookahead benefit.

Reweighting each error by a proxy for its acoustic cost does not rescue it. PER charges every error exactly 1.0, so the natural objection is that the grouping is sound and PER simply mis-weights it: a substituted vowel and a substituted stop cannot sound equally wrong. We test that directly.

Each utterance's canonical phone string is synthesised with a single fixed voice and force-aligned to its own synthesis using an espeak-IPA CTC model, giving a time span per reference phone. The hypothesis string is synthesised with the same voice and DTW-aligned to the reference, and the spectral distortion inside each reference phone's own span is charged to that phone's class. Localisation matters here: a VITS-class synthesiser is globally coupled, so changing one phone perturbs the whole utterance and a whole-utterance distortion cannot be attributed to a single phone. The per-phone span from forced alignment is what makes the measurement local at all.

Synthesis must also be deterministic. Piper defaults to $\mathrm{noise\_scale} = 0.667$ and $\mathrm{noise\_w} = 0.8$, under which the same phone string renders differently on each call: self-distortion is 244.8 against 277.6 for a genuine one-phone change, so roughly 88% of an apparent "cost of an error" would be synthesis noise. Both terms are zeroed, giving bitwise-identical renderings, and the analysis refuses to run otherwise.

Two sanity checks support the measure before it is used, and the verdict is suppressed if validation fails: phones the sequence metric scores as incorrect carry $1.72\times$ the distortion of correct ones (86.8 versus 50.5), and mean distortion falls monotonically with lookahead (71.4 to 46.9, -34%).

The verdict is unchanged (Table VI). Acoustic weighting does move in H2's favour, with effect size $0.607 \to 0.756$ and ordering violations $4/12 \to 2/12$ against the sequence metric on the same utterances, but not far enough to clear the pre-specified threshold of 1.0, and the failure has the same shape as before: the "survives" group's internal spread (0.120) is $3.5\times$ the difference between the groups (0.034), and a nasal again out-gains a diphthong. So this particular equal-cost objection is not supported by the synthesized distortion analysis; it does not exclude weightings this proxy cannot see, notably trajectory errors at correctly identified phones.

TABLE VI
H2 UNDER ACOUSTIC WEIGHTING: RELATIVE REDUCTION IN PER-PHONE DISTORTION, L=0 → 640 MS, 40 UTTERANCES, 1451 REFERENCE PHONES PER CONDITION. B = PRE-SPECIFIED AS BREAKING FIRST, S = AS SURVIVING. THE SEQUENCE-METRIC COLUMN IS COMPUTED ON THE SAME UTTERANCES, SO THE TWO DIFFER ONLY IN WEIGHTING.

	class	acoustic	sequence	phones
B	approximant	0.386	0.825	153
B	vowel (mono)	0.379	0.712	448
S	nasal	0.376	0.784	128
S	fricative	0.372	0.745	272
B	vowel (diph)	0.321	0.667	166
S	stop	0.305	0.663	240
S	affricate	0.257	0.526	44
effect size (need ≥ 1.0)		0.756	0.607	
ordering violations		2/12	4/12	

One explanation does remain open, and it is not the one this test closes. Both signals here are TTS renderings of phone strings, so the synthesiser produces canonical formant trajectories for whatever string it is given. This measures the cost of getting a phone's *identity* wrong; it says nothing about a wrong trajectory for a phone whose identity is right. That still needs real converted audio against a native rendition (§VII). The sample is also far smaller than the sequence-level test, at 1451 unique reference phones per acoustic condition against 42,255 per sequence condition, and is weighted accordingly.

F. RQ3: canonical-phone prediction benefits more from lookahead than realized-phone transcription

Identical architecture, capacity, data order, seed and step count; the two arms differ only in the target tensor. Endpoint gain L=0 → 640 ms is -0.2695 (63.2% relative) for canonical-phone prediction against -0.1893 (42.8%) for realized-phone transcription, a paired difference of $+0.0802$ across the three seeds (sd $= 0.0035$, $t(2) = 39.7$, $p = 6.3 \times 10^{-4}$, 95% CI $[+0.0715, +0.0889]$). With three seeds the interval is wide in principle and narrow here only because the seed-to-seed spread is small relative to the effect.

The dense grid replicates this, and sharpens it. Running the full 16-lookahead × 2-seed sweep on the transcription arm as well puts both arms on the same dense grid (Table VII). The endpoint gains reproduce the 7-point result almost exactly, at 63.3% for canonical-phone prediction against 43.5% for transcription, against 63.2% and 42.8% before, in an independent set of training runs on a denser grid, though on the same corpus and test split. The exchange rates separate cleanly:

$$\text{conversion } -0.0452 \text{ vs transcription } -0.0292 \text{ PER per doubling,}$$

$R^2 = 0.983$ and 0.984 respectively, so **conversion buys $1.55\times$ as much per doubling of lookahead**.

The effect is clearer in the between-arm gap than in the absolute levels. At L=0 the two arms have similar PER (0.4256 vs 0.4427, a gap of 0.0171); by L=640 ms the gap is

0.0939, **5.5× wider**. It widens in 13 of 15 adjacent steps, and the transcription arm is worse at 16 of 16 lookaheads, every one by more than the 0.0029 floor.

The arms differ in their target tensor, which carries more than the task: label entropy, sequence length and annotation noise move with it. We therefore repeat the test within a single canonical-phone model, holding the target representation and the model fixed; the two position subsets can of course still differ in composition and baseline difficulty, which is what the controls below address.

Defining a canonical-deviation position. The `g2p` and `ipa` strings are not positionally aligned and differ by substitutions, deletions and insertions, so a position-i comparison is not available. We align them with Levenshtein backtrace under unit substitution, insertion and deletion costs, and index every quantity by the *canonical* (`g2p`) sequence. A position is a canonical-deviation position when its aligned operation is a substitution or a deletion; an insertion in the `ipa` stream consumes no canonical position and is skipped. Model errors are charged to canonical positions by the same rule, aligning `g2p` against the CTC hypothesis, so that deviation status and error status index the same sequence. Where ties occur the backtrace prefers substitution over deletion over insertion, making the mapping deterministic. One consequence belongs with the result: because everything is indexed to canonical positions, realized-phone insertions are excluded from this analysis, so it covers the substitution and deletion deviations rather than every way the two strings can differ. This is the convention used throughout the paper, including the manner-class analysis of §V-E.

Splitting canonical positions this way, the exchange rate is -0.0520 PER per doubling at canonical-deviation positions against -0.0394 at matching ones, a factor of $\mathbf{1.32}$ from the same model on the same utterances. That converges with the $1.55\times$ between arms. One caveat belongs with it: in *relative* terms the ordering reverses (51% against 71%), because canonical-deviation positions start far harder (0.590 against 0.354) and the gap between the two widens with lookahead. Absolute PER decreases faster at canonical-deviation positions, although these positions begin substantially harder and show a smaller relative reduction.

The interaction survives two controls. Those positions are not a random subset: they differ in baseline difficulty, phone composition and distribution over speakers. Treating 42,255 phone tokens nested in six speakers as independent would substantially overstate precision by ignoring within-speaker dependence, so we bootstrap over *speakers*: the slope difference is -0.0127 with a 95% CI of $[-0.0211, -0.0069]$, excluding zero, and deviation positions are steeper in 2000 of 2000 resamples. Leave-one-speaker-out agrees and is not asymptotic: dropping each talker in turn gives -0.0152 to -0.0095, negative in all six cases and never approaching zero. The result is also insensitive to how ambiguous alignments are resolved: re-running with the backtrace preferring deletion over substitution moves the slope difference from -0.0127 to -0.0131 (ratio 1.32 to 1.33, CI $[-0.0218, -0.0071]$, still 26 of 35 phones in the same direction), so a deterministic tie rule is not doing the work. Recomputing the difference *within each reference phone* and pooling gives -0.0098, steeper in 26 of 35 phones, so phone composition accounts for roughly a quarter of the raw gap but not the effect. We report this as an association under matched conditions, not as evidence that `g2p`≠`ipa` positions are exclusively where accent resides: such positions also include pronunciation variants, reductions and annotation error.

TABLE VII

BOTH ARMS ON THE DENSE GRID, MEAN OF 2 SEEDS. THE GAP IS THE TRANSCRIPTION ARM'S PER MINUS THE CANONICAL-PHONE ARM'S: NEAR ZERO AT L=0 AND SUBSTANTIALLY WIDER AT LARGE LOOKAHEAD; THE FULL 16-POINT CURVE WIDENS IN 13 OF 15 ADJACENT STEPS RATHER THAN MONOTONICALLY.

L (ms)	canonical	transcription	gap
0	0.4256	0.4427	+0.0171
20	0.3700	0.3896	+0.0196
40	0.3275	0.3600	+0.0325
80	0.2754	0.3271	+0.0517
160	0.2206	0.2908	+0.0702
240	0.1973	0.2755	+0.0782
320	0.1823	0.2666	+0.0843
640	0.1561	0.2501	+0.0939
exchange rate (PER/doubling)			
canonical	−0.0452	$R^2 = 0.983$	
transcription	−0.0292	$R^2 = 0.984$	

One secondary point falls out: neither arm yields a locatable knee (breakpoints move to $[40, 240, 160]$ ms across axis treatments for transcription, against $[40, 180, 180]$ for canonical-phone prediction), so §V-C's structural conclusion is not an artefact of the canonical-phone arm. We do not compare saturation onsets between the arms: the five budgets added over 200–360 ms to locate it were run on the canonical-phone arm only, so the transcription arm lacks the resolution the comparison would need. The transcription arm's own curve is just as log-linear, with a shallower slope.

A complementary within-run measure is the *preference margin*, PER against the other arm's labels minus PER against its own. It grows monotonically in 3/3 seeds for the canonical-phone arm ($+0.032$ at L=0 to $+0.099$ at 640 ms; 18/18 adjacent step × seed comparisons positive) while the transcription control is 0/3 monotone and drifts *down* (-0.0103). Building the claim on a difference rather than a level turned out to matter (§VI).

G. t_{buf}: jitter measured, I/O not

t_{buf} splits into a jitter term and an I/O term. The first is measured; the second resisted three separate attempts, and we report that rather than substitute a number.

On the M4 test platform, measured jitter is negligible relative to the other latency terms. Two independent 30 s duplex runs on the M4 at 40 ms blocks give required jitter buffers of 0.14 and 0.08 ms, p99 callback lateness ≈0.06 ms, zero over- or underruns, and clock drift below $1.3 \times 10^{-4}\%$. On this platform the jitter safety margin contributes nothing meaningful to the budget.

The I/O term is unmeasured, and the reason is instructive. An acoustic loopback (emit a chirp, record it, crosscorrelate) failed on every attempt, with all 20 repetitions rejected at correlation 0.042–0.046 against a 0.20 threshold. The diagnosis is not weak signal: captured level is *invariant* to output amplitude (peak 0.00505 at amplitude 0.50 versus 0.00468 at 0.95), so nearly doubling the output changed the captured signal by nothing. That behaviour is consistent with platform echo cancellation, which scales with its own reference and therefore cannot be defeated by a louder probe.

The driver-reported figure is not a substitute. Its excess over the requested block is *identically* 201.69 ms at both 10 and 20 ms blocks before roughly doubling at 40 ms, which strongly suggests the reported quantity includes queueing rather than isolated converter delay: a quantity that is invariant to a doubling of the block size is not measuring the block.

A digital loopback through a virtual audio device, where the signal never becomes sound and echo cancellation cannot reach it, also fails, for a different and more interesting reason. The path works mechanically, but the returned signal peaks near 0.014; lowering the detection threshold until repetitions pass yields p50 206 ms, p90 491 ms, $\sigma = 117$ ms. A distribution whose p90 is $2.4\times$ its p50, on a signal this weak and with this much threshold dependence, is not consistent with a stable measurement of converter latency. We therefore reject these measurements as threshold-dependent and not representative of converter latency: what the procedure records is that a detection threshold was lowered until the tool produced output. Such a device would also measure its own buffering rather than a converter. Closing this term needs a wired loopback through real converters, or a platform without echo cancellation in the path.

VI. Measurement audits and failure modes

A padding bug rewrote the curve's shape. A masked block handed a zero-padded hidden-state tensor to a depthwise convolution and feed-forward that do not respect the key-padding mask. The correction was not a constant offset: it grows with lookahead, from -0.014 PER at $L{=}0$ to -0.086 at 320 ms, $16.4\times$ the noise floor. The buggy curve therefore *understated* the benefit of lookahead, biasing the central quantity of RQ1. A constant offset would have cancelled in every within-run comparison; this did not. Every number reported in this paper was regenerated after the fix.

A forced one-breakpoint estimator returns a breakpoint even when no localised knee exists. ΔBIC cannot distinguish "there is a knee" from "piecewise fits a curve better than a line". Identifiability has to be tested by perturbing the axis and bootstrapping the location, not inferred from ΔBIC alone. Our synthetic check reproduces the artefact exactly: on a curve that is log-linear for $L \geq 20$ with $L{=}0$ off the extrapolation, including $L{=}0$ yields $\Delta\mathrm{BIC} = -39$ and a confident "knee at 40 ms", while dropping it yields $+5$ and no knee at all. An earlier maximum-distance-to-chord estimator reported "knee at 160 ms, bootstrap 95% CI $[160, 160]$" on this data, a tight interval around a plausible value, tight precisely *because* the curve is smooth and every resample of a log-linear curve is also log-linear.

An analysis that buckets unmatched symbols into "other" will produce confident output on the wrong alphabet. Our phoneme-class map was keyed on ARPAbet; the sweep emits IPA. Applying one to the other sent 57% of reference tokens, including *every* vowel and /ɪ/, to "other", and still printed per-class slopes, an H2 verdict and a by-L1 table. Nothing errored. The guard is to assert coverage rather than infer it from the absence of an exception, which the analysis now does.

A fourth attempt, a TTS-counterfactual substitute for forced alignment, was built and rejected before producing any reported number; it is documented in the repository.

VII. Limitations

Every quality result here is phone-level, not perceptual, and this is a phone-translation probe rather than an end-to-end FAC system: the latency–accuracy relation reported here need not transfer to synthesis, prosody, speaker similarity or perceived accentedness. We do not conduct a perceptual study because the phone-sequence outputs are not valid FAC stimuli: synthesized from bare phone strings, they lack the lexical, prosodic and acoustic structure required to judge naturalness or accentedness. This is a property of the stimuli rather than of the budget. Training is 1200 steps at fixed budget over a frozen encoder, a comparison at matched budget rather than at convergence, though §V-B shows the ranking and exchange rate hold to $5\times$ that budget; the dense sweep has two seeds per arm. Its repeatability threshold now comes from a deliberate four-condition $\times$ four-repeat experiment with a bootstrap interval, but that threshold is global while the per-condition σ it pools varies by a factor of 3.5, so marginal decisions near the threshold should be treated as approximate. t_{buf}'s I/O term is unmeasured, and compute is characterised on Apple Silicon and ARM64 Linux only. The encoder comparison in §V-D compares two pretrained encoder checkpoints. The wav2vec 2.0 result is replicated at a second training seed and re-checked at 6000 steps, while the WavLM curve rests on the seeds of the main sweep; that is enough to show the exchange rate does not transfer, but not to attribute the gap to pretraining scale, objective or architecture; and the layer-selection protocols are not fully symmetric: WavLM's layer 9 was fixed a priori from prior practice, whereas wav2vec 2.0's was selected by a phonetic probe at one lookahead, though re-selection at $L{=}0$ returns the same layer.

VIII. Conclusion

For the causal WavLM phone-translation probe, future context reduces PER at an approximately log-linear average rate of $\approx$0.045 per doubling, with no identifiable knee and measurable returns out to a saturation onset near 340 ms, well beyond PHONOS's 40 ms translator budget, relative to which a further 45.3% reduction is available. That rate, however, is measured over one backbone: a second causalised encoder with similar zero-lookahead PER shows only a 3.9% relative endpoint PER reduction from 0 to 640 ms, against 63.3% for WavLM; trained to 6000 steps the corresponding wav2vec reduction is 1.8% (§V-D). Neither the magnitude nor the detailed shape of the lookahead response transfers unchanged across the two encoders tested: WavLM exhibits a strong smooth diminishing-return curve, whereas wav2vec 2.0 shows a weak, non-monotonic one. Future-context budgets should therefore be characterised for the deployed representation rather than inherited from another backbone. Increasing lookahead strongly changes algorithmic latency while only weakly affecting measured per-chunk compute over the tested range, though throughput must still satisfy $t_{\mathrm{cmp}} < c$ (Eq. 3) to run at all.

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X. Compliance with ethical standards

This work used only the publicly released L2-ARCTIC corpus and involved no new participant recruitment, interaction or data collection. No listening study was run, and no claim about perceived accentedness or naturalness is made. The author declares no competing interests.

References

[1] G. Zhao, S. Sonsaat, A. Silpachai, I. Lucic, E. Chukharev-Hudilainen, J. Levis, and R. Gutiérrez-Osuna, “L2-ARCTIC: A non-native English speech corpus,” in *Interspeech*, 2018, pp. 2783–2787.

[2] W. Quamer, M.-R. Tseng, G. Nasrallah, and R. Gutiérrez-Osuna, “PHONOS: PHOnetic neutralization for online streaming applications,” *arXiv preprint arXiv:2603.27001*, Mar. 2026.

[3] W. Quamer, M.-R. Tseng, G. Nasrallah, and R. Gutiérrez-Osuna, “TVTSyn: Content-synchronous time-varying timbre for streaming voice conversion and anonymization,” *arXiv preprint arXiv:2602.09389*, Feb. 2026.

[4] W. Quamer and R. Gutiérrez-Osuna, “DarkStream: Real-time speech anonymization with low latency,” *arXiv preprint arXiv:2509.04667*, Sep. 2025.

[5] T.-N. Nguyen, N.-Q. Pham, Ş. Akti, and A. Waibel, “Streaming non-autoregressive model for accent conversion and pronunciation improvement,” in *Interspeech*, 2025, pp. 4163–4167.

[6] Y. Yang, Y. Kartynnik, Y. Li, J. Tang, X. Li, G. Sung, and M. Grundmann, “StreamVC: Real-time low-latency voice conversion,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

[7] Y. Halychanskyi, J. S. Guo, and V. Kindratenko, “Accent conversion: A problem-driven survey of sociolinguistic and technical constraints,” *arXiv preprint arXiv:2604.27281*, Apr. 2026.

[8] N. Kuzmin, S. Liu, K. A. Lee, and E. S. Chng, “Stream-voice-anon: Enhancing utility of real-time speaker anonymization via neural audio codec and language models,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026.

[9] N. Kuzmin, K. A. Lee, and E. S. Chng, “StreamVoiceAnon+: Emotion-preserving streaming speaker anonymization via frame-level acoustic distillation,” *arXiv preprint arXiv:2603.06079*, Mar. 2026.

[10] W. Quamer and R. Gutiérrez-Osuna, “End-to-end streaming model for low-latency speech anonymization,” *arXiv preprint arXiv:2406.09277*, Jun. 2024.

[11] G. Zhao, S. Ding, and R. Gutiérrez-Osuna, “Converting foreign accent speech without a reference,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 2367–2381, 2021.

[12] S. Ding, G. Zhao, and R. Gutiérrez-Osuna, “Accentron: Foreign accent conversion to arbitrary non-native speakers using zero-shot learning,” *Computer Speech & Language*, vol. 72, p. 101302, 2022.

[13] K. Sadov, M. Hutter, and A. Near, “Low-latency real-time voice conversion on CPU,” *arXiv preprint arXiv:2311.00873*, Nov. 2023.

[14] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, M. Zeng, X. Yu, and F. Wei, “WavLM: Large-scale self-supervised pre-training for full stack speech processing,” *IEEE Journal of Selected Topics in Signal Processing*, 2022.

[15] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[16] R. E. Kass and A. E. Raftery, “Bayes factors,” *Journal of the American Statistical Association*, vol. 90, no. 430, pp. 773–795, 1995.

[17] G. E. Peterson and I. Lehiste, “Duration of syllable nuclei in English,” *The Journal of the Acoustical Society of America*, vol. 32, no. 6, pp. 693–703, 1960.

[18] J. Hillenbrand, L. A. Getty, M. J. Clark, and K. Wheeler, “Acoustic characteristics of American English vowels,” *The Journal of the Acoustical Society of America*, vol. 97, no. 5, pp. 3099–3111, 1995.

[19] T. M. Nearey and P. F. Assmann, “Modeling the role of inherent spectral change in vowel identification,” *The Journal of the Acoustical Society of America*, vol. 80, no. 5, pp. 1297–1308, 1986.

[20] I. Lehiste and G. E. Peterson, “Transitions, glides, and diphthongs,” *The Journal of the Acoustical Society of America*, vol. 33, no. 3, pp. 268–277, 1961.

[21] P. C. Delattre, A. M. Liberman, and F. S. Cooper, “Acoustic loci and transitional cues for consonants,” *The Journal of the Acoustical Society of America*, vol. 27, no. 4, pp. 769–773, 1955.

[22] K. N. Stevens and S. E. Blumstein, “Invariant cues for place of articulation in stop consonants,” *The Journal of the Acoustical Society of America*, vol. 64, no. 5, pp. 1358–1368, 1978.